

Optimize pacemaker programming (e.g., increase sensing threshold) to mitigate oversensing

Executive Technology Brief

Domain: rf_wireless

Technology Maturity: EARLY_CONCEPT

Transfer Posture: Research Partnership

Problem: Pacemaker battery depletion requiring surgical replacement

Mechanism: Mechanically mediated device-device interaction causing oversensing and pacing inhibition

Architecture: PROPOSED architecture (no physical system exists): transmit chain + power limits; channel / tissue path (path loss model); receive chain + detection; regulatory exposure constraint (SAR)

Governing Model: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

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{'equation_id': 'RF-001', 'expression': 'P_r = P_t * G_t * G_r * (lambda / (4 * pi * d))^2', 'source': {'text': 'Friis transmission equation (standard antenna-theory relation)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'far field, line of sight; tissue path requires additional loss term', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['in-body path loss exceeds free-space - margin is MODEL_DERIVED'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'reason': 'domain template rf_wireless declares this relation as part of its governing model', 'mechanism_mentions': []}}
```

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{'equation_id': 'RF-002', 'expression': 'SNR = P_signal / P_noise', 'source': {'text': 'standard detection-theory quantity (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'defined for the chosen detection bandwidth', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['noise figure and interference terms must be budgeted explicitly'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'reason': 'domain template rf_wireless declares this relation as part of its governing model', 'mechanism_mentions': []}}
```

Kill Condition: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

Current State: EARLY_CONCEPT. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.